

Let θ be measured from the downward vertical. The radial (outward) equation of motion is

$$T - Mg \cos \theta = -\frac{Mv^2}{L} \implies T = M\frac{v^2}{L} + Mg \cos \theta,$$

where $T \geq 0$ as long as the string is taut. Conservation of energy gives

$$\frac{1}{2}Mv^2 + MgL(1 - \cos \theta) = \frac{1}{2}Mv_0^2 \implies v^2 = v_0^2 - 2gL(1 - \cos \theta).$$

Q1. Over the horizontal

The string goes slack when $T = 0$:

$$\frac{v^2}{L} + g \cos \theta = 0 \implies v^2 = -gL \cos \theta.$$

Combining with energy,

$$-gL \cos \theta = v_0^2 - 2gL(1 - \cos \theta) \implies v_0^2 = gL(2 - 3 \cos \theta).$$

The slack angle θ' therefore satisfies $\cos \theta' = (2 - v_0^2/gL)/3$. The condition $\theta' > \pi/2$ means $\cos \theta' < 0$, i.e. $v_0^2 > 2gL$. Hence

$$v_{0,\min} = \sqrt{2gL}.$$

Q2. Over the top

The condition $\theta' > \pi$ means $\cos \theta' < -1$ requires $v_0^2 > 5gL$, so the boundary is $\theta' = \pi$, i.e. $\cos \theta' = -1$. Then

$$v_0^2 = gL(2 + 3) = 5gL \implies v_{0,\min} = \sqrt{5gL}.$$

At this speed the string goes slack exactly at the top, $\theta' = \pi$.

Q3. The point of no return

Substituting the energy relation $v^2 = v_0^2 - 2gL(1 - \cos \theta)$ into the tension gives

$$T(\theta) = M\frac{v^2}{L} + Mg \cos \theta = \frac{M}{L}v_0^2 + Mg(3 \cos \theta - 2).$$

Since $\cos \theta$ is minimised at $\theta = \pi$, the tension is minimised at the top:

$$T_{\min} = T(\pi) = \frac{M}{L}(v_0^2 - 5gL).$$

Therefore, if the string is still taut at $\theta = \pi$ (so $T(\pi) > 0$), then $T(\theta) \geq T(\pi) > 0$ for every θ , and in particular for every $\theta > \pi$. So $\theta^* = \pi$ has the required property.

It is also the *minimum*: for any $\theta^* < \pi$, choose a trajectory whose slack angle satisfies $\theta^* < \theta' < \pi$ (such trajectories exist, from Q1 with $\cos \theta' = (2 - v_0^2/gL)/3$). Then $T(\theta^*) > 0$ yet $T(\theta') = 0$, so the property fails. Hence no $\theta^* < \pi$ works, and

$$\theta^* = \pi.$$

Equivalently: the string remains taut for the whole revolution precisely when $v_0^2 \geq 5gL$, and the top is the last point at which it can first go slack.